

QuadSync Tech Solutions

Synchronizing Innovations with Success

BECOME JOB READY IN 90 DAYS

Industry-Oriented Structured Training Program

Data Analyst

(12 weeks)

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Excel :**(Week 1 & 2)**

Chapter No	Topic Name
Chapter 1.	Excel Basics – Interface, Ribbon, Shortcuts, Data Entry, Cell Referencing, Basic Formulas (SUM, AVERAGE, COUNT, MIN, MAX)
Chapter 2.	Formatting & Excel Essentials – Cell Formatting, Conditional Formatting, Data Validation, Sort, Filter & Tables
Chapter 3.	Data Cleaning – Remove Duplicates, TRIM, CLEAN, Flash Fill, Text to Columns, Find & Replace
Chapter 4.	Logical & Text Functions – IF, IFS, AND, OR, IFERROR, LEFT, RIGHT, MID, LEN, CONCAT & TEXTJOIN
Chapter 5.	Statistical, Date & Lookup Functions – COUNTIF, SUMIF, AVERAGEIF, RANK, TODAY, DATEDIF, VLOOKUP, XLOOKUP, INDEX & MATCH
Chapter 6.	Pivot Tables & Dashboards – Pivot Tables, Pivot Charts, Grouping, Slicers, Timelines, KPI Cards, Interactive Dashboards & Report Design
Chapter 7.	Excel with AI – Microsoft Copilot, Analyze Data, AI Formula Generation, ChatGPT for Excel & AI Dashboards
Chapter 8.	Project - Based on Excel

SQL :**(Week 3 & 4)**

Chapter No	Topic Name
Chapter 1.	SQL Basics – Introduction, DBMS vs RDBMS, MySQL Setup, Creating Databases & Tables
Chapter 2.	CRUD Operations – CREATE, INSERT, SELECT, UPDATE & DELETE
Chapter 3.	Keys & Constraints – Primary Key, Foreign Key, Unique, NOT NULL, CHECK, DEFAULT & ALTER
Chapter 4.	Filtering & Sorting – WHERE, Operators, ORDER BY, LIMIT, DISTINCT & Aliases
Chapter 5.	SQL Functions – String, Numeric, Date, Aggregate Functions & GROUP BY, HAVING
Chapter 6.	CASE Expressions & Conditional Queries
Chapter 7.	Joins – INNER, LEFT, RIGHT, FULL, SELF & CROSS JOIN
Chapter 8.	Window Functions – ROW_NUMBER(), RANK(), DENSE_RANK(), LEAD(), LAG(), PARTITION BY
Chapter 9.	Set Operations & Advanced Queries – UNION, UNION ALL, INTERSECT, EXCEPT, Subqueries & CTEs
Chapter 10.	Pivot & Unpivot, Views, Indexes & Query Optimization Basics
Chapter 11.	SQL with AI – ChatGPT for SQL, AI Query Generation, Query Optimization, AI Data Analysis
Chapter 12.	Project – Based on Excel

Python Programming Fundamentals: (Week 5-8)

Chapter No	Topic Name
Chapter 1.	Python Basics – Installation, Variables, Data Types, Input/Output & Basic Syntax
Chapter 2.	Operators, User Input, Type Casting, Expressions & Decision Making
Chapter 3.	Conditional Statements – if, elif, else, Nested Conditions & Logical Operators
Chapter 4.	Loops – for, while, break, continue, Nested Loops & Loop Control
Chapter 5.	Strings – String Methods, Indexing, Slicing, Formatting & String Operations
Chapter 6.	Collections – Lists, Tuples, Sets, Dictionaries, Modules & Built-in Functions
Chapter 7.	Functions – User-Defined Functions, Lambda, Recursion, Scope & Function Arguments
Chapter 8.	File Handling, Exception Handling, Packages & Working with Files
Chapter 9.	Object-Oriented Programming – Classes, Objects, Inheritance, Polymorphism & Virtual Environments
Chapter 10.	Project – Build a Real-World Python Application using Core Python Concepts

Python for Data Analytics (EDA), Visualization & AI:

Chapter No	Topic Name
Chapter 1.	NumPy Basics – Arrays, Indexing, Slicing, Reshaping & Mathematical Operations
Chapter 2.	NumPy Advanced – Broadcasting, Aggregation, Statistics, Random Module & Linear Algebra
Chapter 3.	Pandas Basics – Series, DataFrames, Import/Export, Indexing & Selection
Chapter 4.	Pandas Advanced – Data Cleaning, Missing Values, Filtering, GroupBy, Merge & Pivot Tables
Chapter 5.	Matplotlib – Line, Bar, Pie, Scatter, Histogram & Chart Customization
Chapter 6.	Seaborn – Statistical Plots, Heatmaps, Pair Plots, Box Plots & Correlation Analysis
Chapter 7.	Exploratory Data Analysis (EDA) – Data Cleaning, Descriptive Statistics, Outlier Detection & Feature Analysis
Chapter 8.	Data Visualization – Dashboards, Interactive Charts, KPI Visualization & Data Storytelling
Chapter 9.	Python with AI – ChatGPT for Python, AI Code Generation, Prompt Engineering & AI Insights
Chapter 10.	Project – Build a Real-World Data Analytics Project using Python, NumPy, Pandas, Matplotlib, Seaborn & AI

PowerBI:

Chapter No	Topic Name
Chapter 1.	Power BI Basics – Installation, Interface, Data Sources, Import Data & Power BI Workflow
Chapter 2.	Data Transformation – Power Query Editor, Data Cleaning, Shaping, Merging & Appending Data
Chapter 3.	Formatting & Visuals – Report Formatting, Themes, Tables, Cards & Basic Visualizations
Chapter 4.	Charts & Visualizations – Waterfall, Funnel, Scatter, Treemap, Gauge, Line, Pie, Maps, Matrix, KPI Cards & Custom Visuals
Chapter 5.	DAX Fundamentals – Calculated Columns, Measures, Variables & Common DAX Functions
Chapter 6.	Advanced DAX – CALCULATE, FILTER, Time Intelligence, Ranking & Context Functions
Chapter 7.	Data Modeling – Relationships, Star Schema, Hierarchies, Drill Down, Drill Through & Tooltips
Chapter 8.	Interactive Dashboards – Slicers, Bookmarks, Selections, Navigation Buttons, Report Design & Dashboard Optimization
Chapter 9.	Power BI Service – Publishing, Sharing, Workspaces, Scheduled Refresh, Collaboration & Row-Level Security (RLS)
Chapter 10.	Project – Build an Interactive Dashboard using Power BI, DAX & Power BI Service

Tableau:

Chapter No	Topic Name
Chapter 1.	Tableau Basics – Installation, Interface, Data Connections, Data Sources, Worksheets, Dashboards & Data Preparation
Chapter 2.	Charts & Visualizations – Bar, Line, Pie, Maps, Treemaps, Scatter Plots, Highlight Tables & Chart Formatting
Chapter 3.	Data Analysis – Calculated Fields, Filters, Parameters, Groups, Sets, Hierarchies & Table Calculations
Chapter 4.	Dashboards & Storytelling – Interactive Dashboards, Dashboard Actions, Story Points, Navigation, KPI Design & Best Practices
Chapter 5.	Project – Build a Real-World Interactive Business Dashboard using Tableau with Data Storytelling